

Defining an AI Agent

The question

"AI is the study of rational
agents."

Russell, Stuart J., and Peter Norvig. *Artificial Intelligence: A Modern Approach*.
4th ed., Global ed. Harlow: Pearson, 2022.

Artificial Intelligence
A Modern Approach
Fourth Edition
Global Edition

Stuart J. Russell and Peter Norvig

Contributing writers:

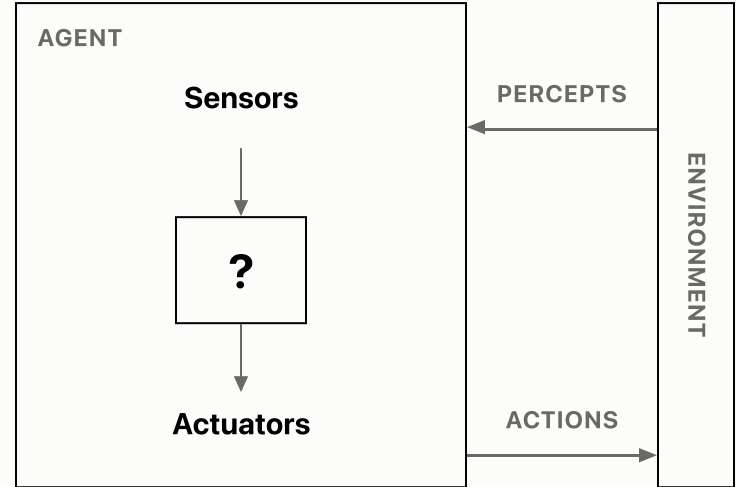
Ming-Wei Chang
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The question

"An agent is anything that can be viewed as perceiving its environment through **sensors** and acting upon that environment through **actuators**."

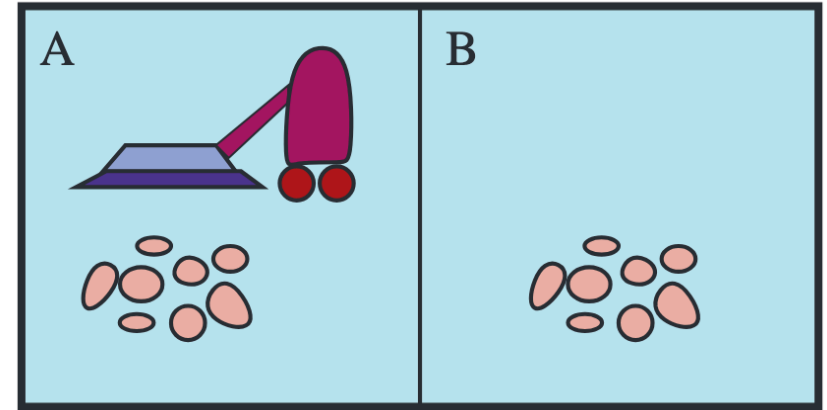
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Percept sequence	Action
[A, Clean]	Right
[A, Dirty]	Suck
[B, Clean]	Left
[B, Dirty]	Suck
[A, Clean], [A, Clean]	Right
[A, Clean], [A, Dirty]	Suck
⋮	⋮
[A, Clean], [A, Clean], [A, Clean]	Right
[A, Clean], [A, Clean], [A, Dirty]	Suck
⋮	⋮

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Mathematically speaking, we say that an agent's behavior is described by the agent function that maps any given agent function percept sequence to an action.

```
function REFLEX-VACUUM-AGENT([location,status]) returns an action
  if status = Dirty then return Suck
  else if location = A then return Right
  else if location = B then return Left
```

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Rational agent

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The nature of environments

Performance measure

Environment

Actuators

Sensors

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits, minimize impact on other road users	Roads, other traffic, police, pedestrians, customers, weather	Steering, accelerator, brake, signal, horn, display, speech	Cameras, radar, speedometer, GPS, engine sensors, accelerometer, microphones, touchscreen

DESCRIPTION OF THE TASK ENVIRONMENT FOR AN AUTOMATED TAXI DRIVER

The nature of environments

"In designing an agent, the first step must always be to specify the task environment as fully as possible."

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits, minimize impact on other road users	Roads, other traffic, police, pedestrians, customers, weather	Steering, accelerator, brake, signal, horn, display, speech	Cameras, radar, speedometer, GPS, engine sensors, accelerometer, microphones, touchscreen

DESCRIPTION OF THE TASK ENVIRONMENT FOR AN AUTOMATED TAXI DRIVER

Properties of task environment

-
- | | |
|----|--|
| 01 | Is it fully or partially observable? |
| 02 | How many agents operate in it? |
| 03 | Is the next state of the environment completely determined by the current state and the action executed? |
| 04 | How does the agent experience the environment - episodically or sequentially? |
| 05 | Can the environment change while an agent is deliberating? |
| 06 | Does the environment has a finite number of distinct states (like in chess game)? |
| 07 | Does the agent know the laws of this environment? |
-

Properties of task environment

THE QUESTION

THE KIND OF ENVIRONMENT

What can the agent see?

Fully observable — partially observable

How many agents act in it?

Single-agent — multi-agent

Does an action fix the next state?

Deterministic — non-deterministic

How does the agent experience it?

Episodic — sequential

Does it change while the agent thinks?

Static — dynamic

How are states and time measured?

Discrete — continuous

Does the agent know how it works?

Known — unknown

The structure of an agent

"agent = architecture + program"

Architecture

The compute, the sensors, the actuators.

Program

The function mapping percepts to actions.

"The job of AI is to design an **agent program** that implements the agent function — the mapping from percepts to actions. We assume this program will run on some sort of computing device with physical sensors and actuators — we call this the **agent architecture**."

Russell, Stuart J., and Peter Norvig. *Artificial Intelligence: A Modern Approach*. 4th ed., Global ed. Harlow: Pearson, 2022.

Types of agent programs

01 **Simple reflex**

02 **Model-based reflex**

03 **Goal-based**

04 **Utility-based**

What we got so far

01 An agent perceives and acts in an environment. Its **agent function** maps any percept sequence to an action.

02 The **performance measure** judges behaviour. A rational agent maximises its expected value, given what it has perceived.

03 A **task environment** = performance measure + environment + actuators + sensors. Specify it first.

04 Environments vary: observable or partial, single- or multi-agent, deterministic or not, episodic or sequential, static or dynamic, discrete or continuous, known or unknown.

05 If the performance measure is unclear, the agent optimises the wrong objective. **The agent design should reflect uncertainty about the true objective.**

06 The **agent program** implements the agent function. Designs differ in efficiency, compactness and flexibility — the environment decides which fits.

07 Reflex → model-based reflex → goal-based → utility-based: more internal state, more flexibility.

08 All agents can improve their performance through **learning**.

Wooldridge & Jennings (1995): the weak notion of agency

Autonomy

Acts without direct human intervention; controls its own state.

Reactivity

Perceives changes in the environment and responds in time.

Pro-activeness

Takes initiative toward goals, not just responds.

Social ability

Interacts with other agents, and with humans.

The strong notion of agency

"...conceptualised or implemented using concepts that are more usually applied to humans — knowledge, belief, intention, obligation."

Wooldridge, Michael, and Nicholas R. Jennings. "Intelligent Agents: Theory and Practice." *The Knowledge Engineering Review* 10, no. 2 (1995): 115–52. <https://doi.org/10.1017/S0269888900008122>.

Franklin & Graesser (1996): temporal closure

"An autonomous agent is a system situated within and a part of an environment that senses that environment and acts on it, over time, in pursuit of its own agenda and so as to effect what it senses in the future."



Then: LLMs happen

2017	Transformer (Vaswani et al.)
2020	GPT-3
2022	Instruction-following LLMs at scale

The concept of Chain-of-thought

"We explore how generating a chain of thought—a series of intermediate reasoning steps—significantly improves the ability of large language models to perform complex reasoning."

Wei, Jason, Xuezhi Wang, Dale Schuurmans, et al. "Chain-of-Thought Prompting Elicits Reasoning in Large Language Models." arXiv:2201.11903. Preprint, January 10, 2023.
<https://doi.org/10.48550/arXiv.2201.11903>.

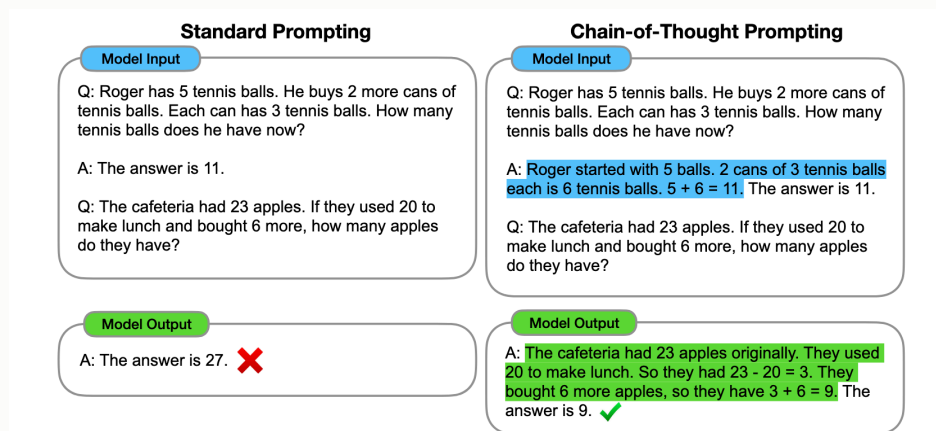


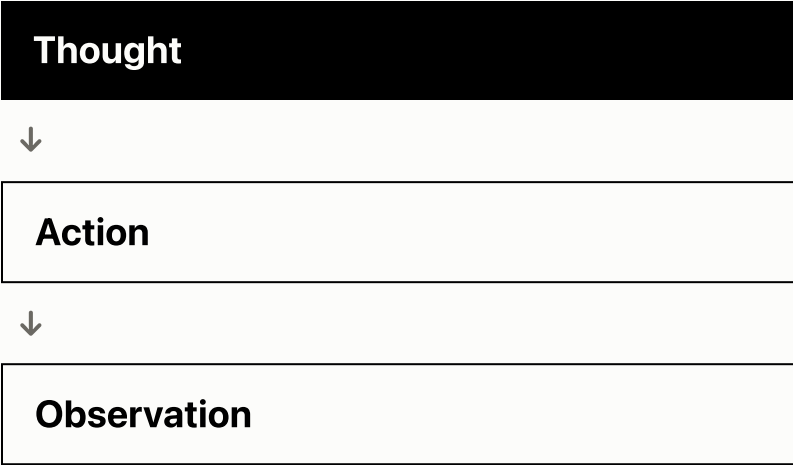
Figure 1: Chain-of-thought prompting enables large language models to tackle complex arithmetic, commonsense, and symbolic reasoning tasks. Chain-of-thought reasoning processes are highlighted.

What happened next

01	2022 CoT prompting	A prompt trick
02	2022 Zero-shot CoT	"Let's think step by step" (Kojima et al.) works with no examples at all, one phrase in the system prompt. Still just prompting.
03	2023 Supervised fine-tuning	Train on tasks paired with reasoning traces written by people or other models. The model learns to imitate the form of step-by-step reasoning.
04	2024–2025 RL on reasoning	o1, DeepSeek-R1. Hard task with a checkable answer → long reasoning trace → verify the final answer automatically → reinforce the traces that worked, weaken the rest → repeat millions of times.

The model stops imitating human reasoning and **discovers which strategies actually reach the right answer.**

The LLM alternates:



↓ back to Thought

Prompt Method ^a	HotpotQA (EM)	Fever (Acc)
Standard	28.7	57.1
CoT (Wei et al., 2022)	29.4	56.3
CoT-SC (Wang et al., 2022a)	33.4	60.4
Act	25.7	58.9
ReAct	27.4	60.9
CoT-SC → ReAct	34.2	64.6
ReAct → CoT-SC	35.1	62.0
Supervised SoTA ^b	67.5	89.5

Table 1: PaLM-540B prompting results on HotpotQA and Fever.

^aHotpotQA EM is 27.1, 28.9, 33.8 for Standard, CoT, CoT-SC in Wang et al. (2022b).

^b(Zhu et al., 2021; Lewis et al., 2020)

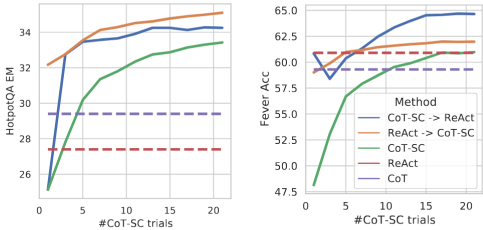


Figure 2: PaLM-540B prompting results with respect to number of CoT-SC samples used.

Yao, Shunyu, Jeffrey Zhao, Dian Yu, et al. "ReAct: Synergizing Reasoning and Acting in Language Models." arXiv:2210.03629. Preprint, March 10, 2023.
<https://doi.org/10.48550/arXiv.2210.03629>.

R&N ↔ ReAct

RUSSELL & NORVIG, 1995

percept → decide → action → ...

Sensors

Agent program

Actuators

REACT, 2023

Observation → Thought → Action → ...

Text input from tools

LLM prompt + generation

Tool calls

Toolformer: tool use is a learnable capability

LLMs can be trained to insert tool calls into their own generation.

```
[CALL calculator(2+2)]
```

Schick, Timo, Jane Dwivedi-Yu, Roberto Dessì, et al. "Toolformer: Language Models Can Teach Themselves to Use Tools." arXiv:2302.04761. Preprint, February 9, 2023. <https://doi.org/10.48550/arXiv.2302.04761>.

The New England Journal of Medicine is a registered trademark of [QA("Who is the publisher of The New England Journal of Medicine?") → Massachusetts Medical Society] the MMS.

Out of 1400 participants, 400 (or [Calculator(400 / 1400) → 0.29] 29%) passed the test.

The name derives from "la tortuga", the Spanish word for [MT("tortuga") → turtle] turtle.

The Brown Act is California's law [WikiSearch("Brown Act") → The Ralph M. Brown Act is an act of the California State Legislature that guarantees the public's right to attend and participate in meetings of local legislative bodies.] that requires legislative bodies, like city councils, to hold their meetings open to the public.

Figure 1: Exemplary predictions of Toolformer. The model autonomously decides to call different APIs (from top to bottom: a question answering system, a calculator, a machine translation system, and a Wikipedia search engine) to obtain information that is useful for completing a piece of text.

The concept of memory

01	Cross-episode learning	Reflexion (Shinn, March 2023). Problem: The agent fails, you restart it — and it has no idea what it already tried. After a failed attempt the agent writes a verbal reflection into a separate buffer, and the next attempt reads it. The first explicit split: short-term trace vs long-term reflections .
02	Long simulations	Generative Agents (Park, April 2023). A memory stream with retrieval by recency + importance + relevance, reflection into higher-level summaries, and planning. Here episodic memory became a first-class concept.
03	Reusable skills	Voyager (Wang, May 2023). A Minecraft agent with a skill library that accumulates reusable code skills — the first strong demonstration of procedural memory .

In parallel, the RAG literature (Lewis et al., 2020 onwards) had been handling external facts for years.

CoALA: cognitive architecture for language agents

Memory	Working, episodic, semantic, procedural
Action space	Internal (reasoning, memory writes) and external (tools)
Decision procedure	How the policy picks the next action

Sumers, Theodore R., Shunyu Yao, Karthik Narasimhan, and Thomas L. Griffiths. "Cognitive Architectures for Language Agents."

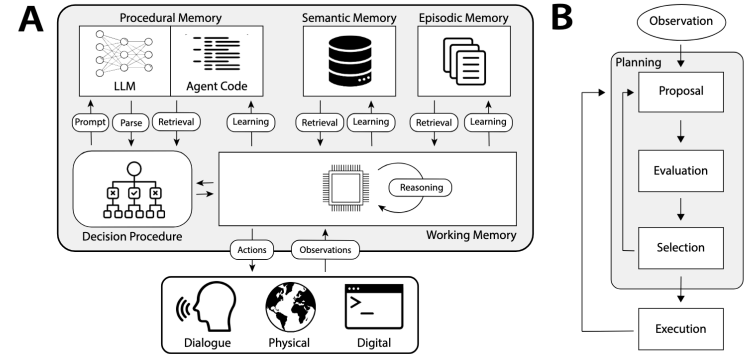


Figure 4: Cognitive architectures for language agents (CoALA). **A:** CoALA defines a set of interacting modules and processes. The **decision procedure** executes the agent's source code. This source code consists of procedures to interact with the LLM (prompt templates and parsers), internal memories (retrieval and learning), and the external environment (grounding). **B:** Temporally, the agent's decision procedure executes a **decision cycle** in a loop with the external environment. During each cycle, the agent uses **retrieval** and **reasoning** to plan by proposing and evaluating candidate **learning** or **grounding** actions. The best action is then selected and executed. An observation may be made, and the cycle begins again.

Agency is frame-dependent

"Agency is not an absolute property of a system. It is a property of the pair (system, reference frame)."

Abel, David, André Barreto, Michael Bowling, and Will Dabney. "Agency Is Frame-Dependent.", 2025

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"Agency Is Frame-Dependent.", 2025

A REFERENCE FRAME

- 1 **Boundary** — what is inside the agent, what is environment
 - 2 **Reference object** — against what we judge that the system caused its action
 - 3 **Goal principle** — what counts as meaningfully pursuing a goal
 - 4 **Adaptation criterion** — which behaviour changes count as adaptation
-

**Do you think AI agents are possible
today?**

Aren't today's "agents" just scaffolding?

Claude Code, Cursor, Operator, AutoGen, even humanoid stacks: **engineering scaffolding around a frozen model.**

Xing, Eric, Mingkai Deng, and Jinyu Hou. "Critique of Agent Model." Preprint, arXiv, 2026.

WHERE THE COMPETENCE LIVES

- 01 The model contributes **reasoning steps** — nothing more
 - 02 Goals, role, planning, stop condition, learning are written **outside**: system prompt, Python loop, pipeline
 - 03 Every new task means an engineer **rewrites the harness**
 - 04 The autonomy belongs to **the builder**, not to the system
-

Agentic vs agentive

COMPONENT	AGENTIC (BOLTED ON)	AGENTIVE (INTERNALIZED)
Goals	Step-by-step instructions in the prompt	Decomposes a long-horizon goal itself
Identity	A fixed system prompt	Maintains and updates a self-model
Decision-making	Unconstrained chain-of-thought	Internal world model, simulated consequences
Self-regulation	Loop depth set by the engineer	Chooses fast, slow, or plan
Learning	Manual retrain–deploy cycle	Learns continuously from deployment

GIC: what an agentic architecture would need

GOAL — IDENTITY — CONFIGURATOR

One reasoning policy $\pi(a | s, g, i)$ — action conditioned on world state, goal and identity

World model trained separately — predicts consequences for simulation; not the policy

Configurator — decides how much reasoning this step deserves

Fast / slow updates — continual learning instead of retrain-deploy

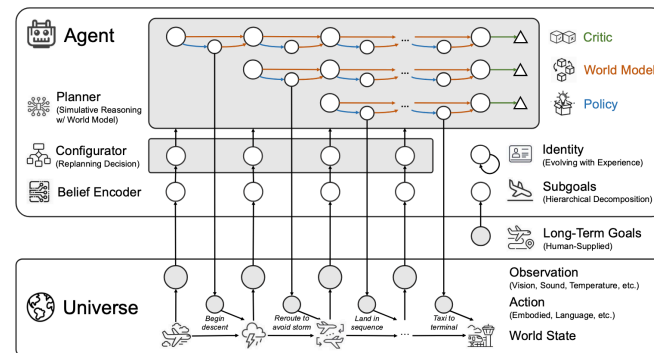
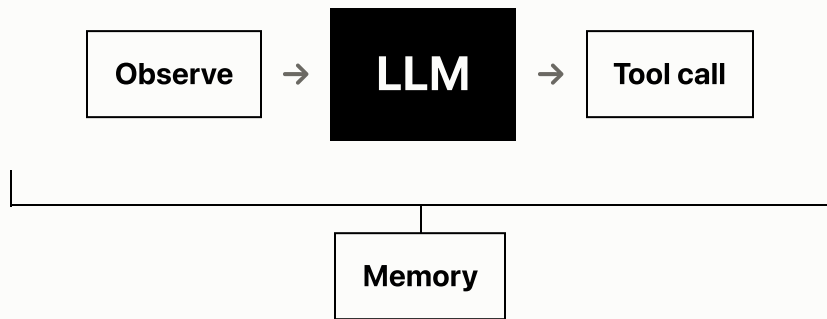


Figure 8: The GIC Agent Model architecture, illustrated with the aircraft pilot use case. **(Bottom)** The universe emits observations and receives actions from the agent. **(Top)** The agent processes observations through a *belief encoder* to form belief states, conditioned on an evolving *identity* and hierarchically decomposed *subgoals*. The *configurator* (System III) decides at each step whether to invoke the planner or act directly. When planning is invoked, the *planner* (System II) simulates candidate trajectories: the *world model* predicts future states, the *policy* proposes candidate actions, and the *critic* evaluates expected long-term value. The best plan is executed through the agent's actions (System I).

What agents we will be building

- 01 **A loop: perceive → decide → act**
- 02 **Sensors. The agent reads observations from its environment**
- 03 **Actuators. The agent acts through tool calls**
- 04 **A decide block powered by an LLM**
- 05 **Memory (adequate to your task)**



NEXT LECTURE → 24.09

Anatomy I: a minimal ReAct agent in code